CHANDU PRASAD BHAIRAPU



GenAI & Agentic Systems • Industrial AI & Data Science

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ABOUT ME

AI engineer (MSc in Artificial Intelligence, 3 years of experience) who takes GenAI and agentic systems from concept to working code, with a spike in industrial data. Sole developer of a multi-agent document intelligence system for the German energy sector at a start-up (precision/recall 33%/45% → 93%/90%), working directly with SAP domain experts and reporting to the CTO; two internships and a master's thesis at Covestro (time-series process data, an agentic AI app on AWS, RAG evaluation over chemical data); studied and worked in India and Germany.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Translated the requirements of stakeholders (SAP domain experts across the German energy sector) into an agentic document intelligence system, built end-to-end as sole developer (data ingestion, chunking, prompt engineering, indexing with agentic, deterministic, and monolithic strategies), reporting directly to the CTO.
- Designed a multi-agent system with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for regulatory and technical documents, raising precision/recall from 33%/45% to 93%/90% against an SAP-system-derived ground truth.
- Engineered a deterministic (code-based) pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, where domain experts review and approve knowledge-graph changes.
- Built a Model Context Protocol (MCP) service giving AI agents millisecond-latency access to the knowledge graph, integrated into Claude Code for agentic development workflows.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, foundation models (LLM APIs), MCP, GraphRAG, RAG, Neo4j, MongoDB, Postgres, FastAPI, React, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows and advised on how agentic AI could support them, translating their requirements into a technical solution approach.
- Designed and deployed the resulting agentic AI application with LangGraph, routing requests across vector database queries, general inference, and summarization.
- Built the full stack (React frontend, Python/FastAPI backend), containerized it with Docker and deployed it to AWS with CDK and a GitLab CI/CD pipeline.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation using CDK.
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, S3, CDK), Azure AI Search, Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from industrial time-series process data, engineering lag features and rolling statistical windows from sensor data.
- Benchmarked the models against empirical curve-fitting and improved early identification of batches likely to deviate from specification, an input to process optimization.
- Built a Streamlit dashboard letting chemists simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Pandas, NumPy, Scikit-Learn, XGBoost, TensorFlow/Keras, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- At a global IT services and consulting company, developed and compared ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for electric vehicle load time-series forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Plant Maintenance Agent | Claude Agent SDK | Microsoft Foundry | Snowflake | dbt

- Multi-agent proof of concept for industrial maintenance: one agent analyses machine sensor time series (OPC UA-style tag model) for anomalies, another searches equipment manuals with document intelligence, and an orchestrator proposes a maintenance action for the plant operator to confirm.
- Sensor and maintenance data modelled with dbt in Snowflake; agents built with the Claude Agent SDK and deployed via Microsoft Foundry on Azure, with an evaluation set run in CI/CD.

TECHNICAL SKILLSET

- **GenAI & Agentic Systems:** multi-agent frameworks (LangGraph, LangChain), foundation models via LLM APIs (LiteLLM), MCP, RAG, GraphRAG, prompt engineering, evaluation (RAGAS, PytrecEval), Claude Code
- **Data Science & Machine Learning:** time-series analysis and forecasting (ARIMA, SARIMA, Prophet, LSTM), Scikit-Learn, XGBoost, PyTorch, TensorFlow, Pandas, NumPy
- **Programming, Cloud & Engineering:** Python, Azure (Azure AI Search), AWS (CDK, Lambda, EC2, SQS, DynamoDB, S3), Docker, Kubernetes, FastAPI, GitLab CI/CD
- **Data:** SQL, Postgres, MongoDB, Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)